

This document outlines the test cases written for the *PaymentProcessor* class, covering payment processing behavior, validation of input, and delivery fee calculations.

Credit Card - First Order

The test method *testCreditCardFirstOrder()* verifies that when a user makes their first payment using a credit card, a 15% discount is applied to the base amount, followed by a 15% tax on the discounted total. For a \$100 payment, the amount after a 15% discount becomes \$85. Applying a 15% tax results in a final amount of \$97.75.

Credit Card - Non-First Order

The test *testCreditCardNonFirstOrder()* checks that returning customers using a credit card receive only a 5% discount. The discounted amount of \$95 is then taxed at 15%, producing a final result of \$109.25.

PayPal - First Order

The *testPaypalFirstOrder()* case tests the behavior when a user pays with PayPal for the first time. A 12% discount is applied, lowering the \$100 amount to \$88. After applying a 15% tax, the final charge becomes \$101.20.

PayPal - Non-First Order

In the *testPaypalNonFirstOrder()* test, returning users paying with PayPal receive a 2% discount. This reduces \$100 to \$98, and the application of a 15% tax brings the final amount to \$112.70.

Cash - First Order

The *testCashFirstOrder()* ensures that new customers paying with cash get a 10% discount. The \$100 amount becomes \$90 after the discount, and the final charge after 15% tax is \$103.50.

Cash - Non-First Order

For non-first-time users paying with cash, the test *testCashNonFirstOrder()* confirms that no discount is applied. The full \$100 amount is taxed at 15%, resulting in a total of \$115.00.

Negative Amount Handling

The *testNegativeAmountThrowsException()* test validates that the system does not allow negative payment amounts. It ensures that an *IllegalArgumentException* is thrown when a payment of -\$50 is attempted.

Zero Amount Handling

Similarly, the *testZeroAmountThrowsException()* checks that zero-value payments are invalid. Attempting to process a \$0 payment triggers an *IllegalArgumentException*.

Delivery Fee - Below Threshold

The *testDeliveryFeeBelowThreshold()* test checks that a \$5 delivery fee is added when the amount is less than \$50. Specifically, for a \$49.99 amount, the system correctly returns a fee of \$5.00.

Delivery Fee - At Threshold

The *testDeliveryFeeAtThreshold()* ensures that no delivery fee is charged when the amount is exactly \$50. The delivery fee in this case is expected to be \$0.00.

Delivery Fee - Above Threshold

Finally, the *testDeliveryFeeAboveThreshold()* validates that no delivery fee is applied to amounts above \$50. For a \$99.99 payment, the delivery fee remains \$0.00.